

Google

LC 2226 - Max Candies Allocated to K children

We have piles which can be divided into subpiles.

Now one child can get candies only from one pile.

Eg 1

[5, 8, 6] | $k = 3$

Ask: Can every child have 4 candies?

- 5 → Support 1 child
- 8 → Support 2 child
- 6 → Support 1 child

Possible

so 3, 2, 1 also possible

Eg 2

Ask: Can every child receive 6 candies?

- $5 \rightarrow 0$
- $8 \rightarrow 1$
- $6 \rightarrow 1$

Only 2 childs

2 < 3 not possible

So hint (MONOTONICITY + max) $\rightarrow$ binary search

`l = 1  $\rightarrow$  1 candy to min child`

`h = *max  $\rightarrow$  since 1 kid can take only from 1 pile`

C++ Solution

```
class Solution {
public:
    bool check(int mid, vector<int>& candies, long long k) {
        long long fed = 0;

        for(int pile : candies) {
            fed += pile / mid;
        }

        if(fed < k) return false;
        else return true;
    }

    int maximumCandies(vector<int>& candies, long long k) {
        int n = candies.size();
        long long tot = accumulate(candies.begin(), candies.end(), 0LL);
        if(k > tot) return 0;

        int l = 1;
        int h = *max_element(candies.begin(), candies.end());

        int ans = 0;
        while(l <= h) {
            int mid = l + (h - l) / 2;
            if(check(mid, candies, k)) {
                ans = mid;
                l = mid + 1;
            } else {
                h = mid - 1;
            }
        }

        return ans;
    }
};
```